

PUPIL WORKSHEET

Earth and the planets

Autumn 1 · Week 6 · Lesson A

Describe Earth and the other planets orbiting the Sun: order eight planets, group rocky/giant planets, and explain the Sun's inward gravitational pull.

Monday · 40 minutes. Name: ____ Date: ____ . Order the planets, classify examples and explain the orbit. Keep this sheet for Tuesday.

1. Arrival: order from the Sun

Choose your route. Supported: order the first four names with an adult. Standard: write those four. Stretch: try all eight. Word bank: Neptune, Mars, Mercury, Jupiter, Earth, Uranus, Venus, Saturn.

2. Watch and explain the orbit model

Write, draw, point or dictate as agreed with your teacher.

Draw a Sun and Earth. Add a gravity arrow pointing towards the Sun and a separate motion arrow along the path. Label what each arrow means. Move Earth to a second position and reset both arrows. Use words or direct an adult if preferred.

3. Rocky or giant?

Write, draw, point or dictate as agreed with your teacher.

Classify each planet as rocky, gas giant or ice giant. Then compare the year values below.

Planet	Rocky / gas giant / ice giant
Mercury	
Mars	
Jupiter	
Neptune	

4. Compare one year

Earth takes about 365 Earth days for one orbit; Mars takes about 687. Which has the longer year? Use the numbers as evidence.

5. Sort and repair

Write, draw, point or dictate as agreed with your teacher.

Mark each original claim true or false. Choose one false claim and explain its correction.

Original claim	True / false
Planets orbit the Sun.	
The Sun orbits Earth.	
Gravity keeps planets in orbit.	
There is no gravity in space.	
Further-out planets have longer years.	

6. Build the explanation

Complete in speech, arrows or writing: The Sun's gravity pulls Earth_____, which continually_____. Keep your work and choose one idea to check on Tuesday.

CHOOSE A RESPONSE ROUTE

Ways to show your understanding

Choose the route agreed with your teacher. You do not need to complete every route.

Supported

Order name cards with an adult, identify Earth, point to the Sunward gravity arrow and compare 365 with 687.

Standard

Order all eight, classify the four examples, label both arrows and explain the Sun's gravitational pull.

Stretch

Explain why the motion arrow is not another force and predict the straight-on tangent if gravity were absent.